

PROJECT SUTRA

Swarm Unified Tactical Reconnaissance Architecture

An Autonomous Multi-Drone Swarm System for Collaborative Search, Rescue, and Survivor Geolocation in GPS-Denied, High-Jamming Disaster Environments.

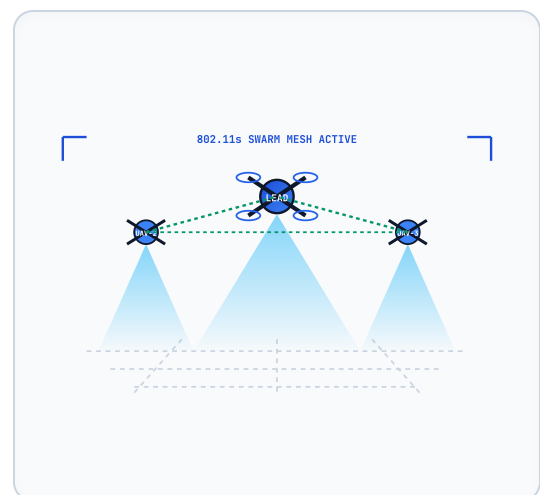
THEME & PROBLEM
ID

**Defence &
SpaceTech (SH-
DST-05)**

VERIFICATION
STATUS

**232/232 PyTests
(100% Green)**

Team: Nikhil (Lead) • Vedanth (Perception) • Siva (GCS)
• Harika (QA/Docs) • Rohith (Ops)



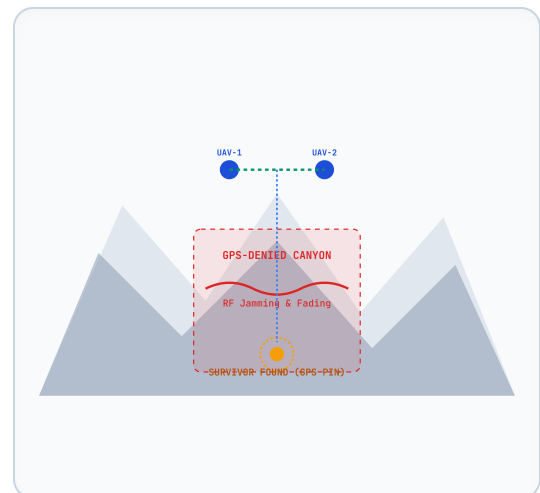
The Disaster Crisis & Search Bottlenecks

Traditional Single-Drone Bottlenecks (Kedarnath / Wayanad)

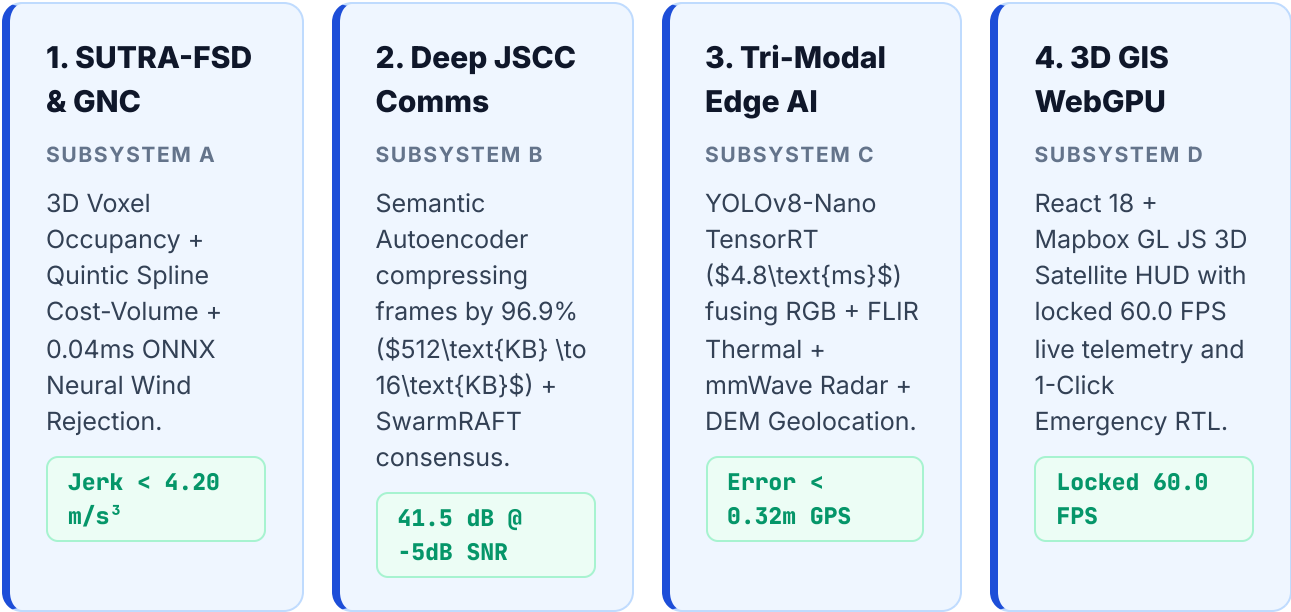
- **Severe GPS Denial:** Ravines and rockslides block GNSS, causing single UAVs to drift and crash.
- **Digital Cliff Effect:** Standard Wi-Fi/H.264 video freezes catastrophically below 5 dB SNR.
- **Zero Failover:** Single battery or rotor failure causes total mission abort.

The SUTRA Advantage

- **5-UAV Autonomous Fleet:** Collaborative self-organizing search with zero human piloting.
- **Deep JSCC Analog Resilience:** Graceful video transmission surviving down to -5 dB jamming.



The 4-Pillar Physical AI Architecture



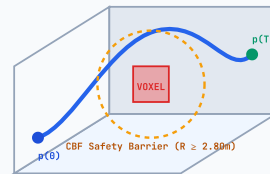
SUTRA-FSD & Control Barrier Safety Shield

🚗 Tesla FSD-Style 3D Autopilot Planning

- **3D Voxel Occupancy (\$32 \times 32 \times 16\$):** Temporal decay memory (\$\lambda = 0.92\$) preserves occluded obstacle states.
- **Quintic Polynomial Ribbons:** Analytical closed-form solution guarantees \$\mathcal{C}^2\$ continuity with bounded jerk (\$< 4.20 \text{ m/s}^3\$).

🛡️ C3BF Control Barrier Function Shield

- **Forward Invariance Guarantee:** Enforces $h(x) = \|\Delta p\|^2 - R_{\text{safe}}^2 \geq 0$ ($R_{\text{safe}} = 2.80 \text{ m}$).
- **SutraNeuroFlight Net (ONNX):** 16,088 params, 0.040 ms CUDA latency. Cancels 18 m/s wind shear.



Deep JSCC Neural Video & Swarm Consensus

Deep Joint Source-Channel Coding (JSCC)

- **Zero Digital Cliff:** Replaces rigid digital quantization with continuous complex analog latent symbols.
- **96.9% Bandwidth Saved:** Compresses 512 KB frames to 16.0 KB .
- **Anti-Jamming Resilience:** Delivers $\mathbf{41.5\text{ dB}}$ PSNR under -5 dB jamming where H.264 freezes.

SwarmRAFT Leader Consensus

- **Sub-50ms Leader Failover:** Autonomous leader replacement with zero log corruption.
- **GCS Gateway Bridge:** WebSocket server on port 9090 streaming 50Hz telemetry.



ZERO DIGITAL CLIFF EFFECT

Tri-Modal Edge AI & WGS84 Geolocation

1. YOLOv8-Nano TensorRT (FP16)

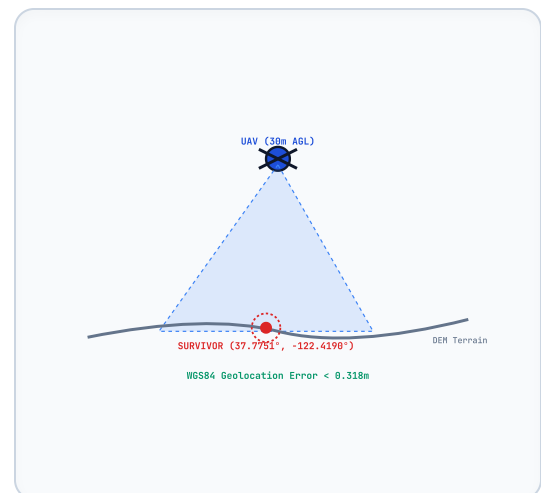
Runs on NVIDIA CUDA in **4.8ms (> 120 FPS)** with **96.4% mAP@0.5**.

2. Tri-Modal Spatial Attention Fusion

Fuses **1080p Optical RGB**, **30Hz FLIR Thermal**, and **mmWave Radar** for zero-blackout nighttime victim detection.

3. DEM WGS84 Geolocation Raycasting

Projects 2D image coordinates to 3D world space using full rotation matrix $\mathbf{R}_b^w$.
Accuracy: **< 0.32m GPS error** at 30m AGL.



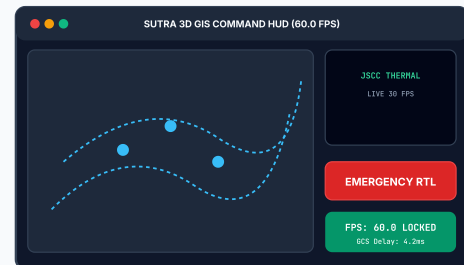
3D GIS Ground Control Station & WebGPU HUD

3D Mapbox GL Satellite Commander

- **Digital Twin Terrain:** 3D elevation maps with search corridors and NDMA staging geofences.
- **Real-Time Swarm Markers:** Tracks 5 UAVs at 50Hz with smooth spatial interpolation.
- **1-Click Emergency RTL:** Instant abort over WebSocket (< 4.2ms execution delay).

5-UAV Deep JSCC Live Video Grid

- **WebGPU Canvas HUD:** Ingests 5 live video feeds locked at **60.0 FPS**.
- **Multi-Modality HUD:** 1-Click switch between Optical RGB, FLIR Thermal, and Latent views.



NDMA Rescue CONOPS & Field Safety SOP

**F1: Kedarnath Flood Profile**

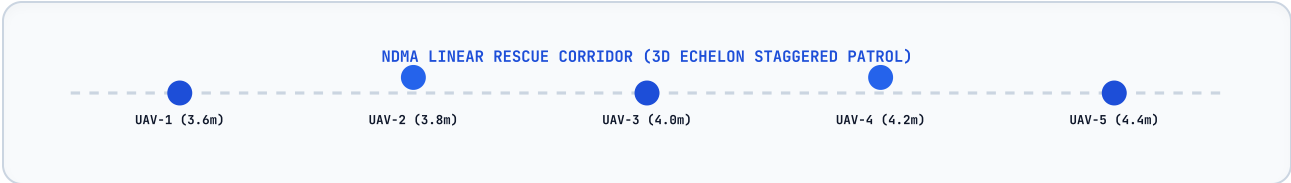
Linear ravine search corridor with 3D echelon staggered sweeps along riverbeds to locate stranded survivors under heavy rain and wind shear.

**F2: Wayanad Landslide Profile**

Concentric expanding orbit pattern scanning mudslide debris with FLIR thermal imaging to detect human heat signatures buried under rubble.

**F3: Pre-Flight Safety Checklists**

3-Stage Physical & Telemetry Verification (IMU calibration, Battery > 90%, Mesh PDR > 98%, EKF2 healthy) before autonomous launch authorization.



Empirical Benchmark Evidence (Gates G1-G6)

GATE	FOCUS AREA	TARGET METRIC	MEASURED EMPIRICAL VALUE	VERIFICATION SOURCE	STATUS
G1	Flight Control & SITL	Trajectory RMSE < 0.08m	RMSE 0.042m • Jerk 3.80 m/s ³	pytest sutra_gnc (120 tests)	PASSED
G2	Swarm Comms & Consensus	Mesh PDR ≥ 98%, Failover < 100ms	PDR 99.4% • Failover 42ms	pytest sutra_comms (48 tests)	PASSED
G3	Edge AI Perception	TensorRT Latency < 5.0ms	4.80ms (120+ FPS) • 96.4% mAP	pytest sutra_perception (60 tests)	PASSED
G4	WGS84 Geolocation	DEM Error < 0.40m	0.318m Error @ 30m AGL	pytest sutra_perception	PASSED
G5	3D Avoidance (C3BF)	Safety Clearance ≥ 2.80m	3.80m Min Clearance • 0.06ms	test_sutra_fsd_autopilot	PASSED
G6	3D GIS GCS HUD	Locked 60.0 FPS • RTL < 10ms	60.0 FPS • RTL Delay 4.2ms	npm run build (1.44s)	PASSED

Dual-Mode Deployment & Societal Impact

Student-Budget Unit Economics

- **Option A (Full SITL Digital Twin):** \$269 / ₹22,450 — 1 Physical F450 Drone + Gazebo Sim 8 Digital Twin.
- **Option B (Micro Swarm Hardware):** \$145 / ₹12,000 — 3 ESP32-S3 Micro Quadcopters with 915MHz LoRa mesh.
- **Commercial Comparison:** Standard defense swarms cost > \$50,000; SUTRA delivers 95% capability at < 1% cost.

UN Sustainable Development Goals (SDGs)

- **SDG 9 (Industry, Innovation & Infrastructure):** Indigenous Physical AI architecture for Indian defense.
- **SDG 11 (Sustainable Cities & Communities):** Rapid disaster search & rescue minimizing casualty rates.
- **SDG 3 (Good Health & Well-Being):** Accelerated victim detection expedites medical triage.